

Algebra

Week 1A

Overview

Introduction to Algebra and Expansion

- Key Ideas*
- *What is Algebra*
 - *Standard Sets of Number*
 - *Substitution*
 - *Simplification*
 - *Expanding using Distributive Law*

Key Idea – What is Algebra

What is Algebra? Algebra – starting with the use of variables to represent numbers is first introduced in early High school. However, sometimes the skills are not learned correctly, or they are not practiced consistently. As all calculus relies on a solid knowledge of Algebra, these skills are important for success in any Mathematics subject at university level.

Here are some examples of algebra that you might have seen at school:

$$2 + 7 = \square \quad 10 - \square = 7 \quad \square \times 5 = 30 \quad 18 \div \square = 9.$$

They are basically little number-puzzles or games (like a sudoku or a crossword). The rules of the game are that you are given an equation with an empty box in it. The objective of the game is to find the number that you have to write in the box to make the equation correct.

That's it. That's the idea of algebra.

We can make harder puzzles by allowing more empty boxes in the equation, and equations that have more operations in them. For example,

$$2 \times \square = \square + 7.$$

Since there are multiple boxes now, we have to be clear about the rules. The rule is that to solve the puzzle, you have to find a single number so that when you write that same number in all the boxes, the equation is correct.

In this example, after a bit of trial and error, you would probably spot that if you write a 7 in each of the two boxes, the equation is correct. You might even have noticed the trick that since $2 \times \square$ is the same thing as $\square + \square$, the puzzle is equivalent to this slightly different puzzle:

$$\square + \square = \square + 7.$$

And it might occur to you, after trying some numbers, that no matter what number you put in the boxes, one of the boxes on the left will “balance out” the box on the right, and the discrepancy between the two sides will be the difference between the number in the second box on the left and the 7 on the right. And that might make you think that 7 would be a good guess for what to put in the boxes. And you would be correct.

It might also occur to you that this was a handy trick you just came up with, and that there might be a good system of handy tricks that would help you get on to the “advanced” sudokus in the back of the book. And you would be correct about that too. And you’d have begun the process of re-inventing algebra.

As you got better and better at the puzzles, you would probably go looking for harder puzzle books. The puzzle setters would start by making the expressions more complicated:

$$2 \times \square = \square + 7 \square + 2 \div 7 + 5 = 23 - \square.$$

Then they might start to include other operations to try to make the puzzles harder:

$$2^{\square} + 7 = 71.$$

Eventually, to make it tricky, they might introduce multiple types of boxes and multiple equations with new rules:

$$2 \times \bigcirc + \square - 7 = 3 \times \square - \bigcirc$$

$$\bigcirc + 3 \times \square = \square - \bigcirc$$

(1) all boxes of the same shape have to contain the same number across all equations, but boxes of different shapes contain different numbers.

(2) to solve the puzzle you must make ensure that all the equations are correct simultaneously.

What we teach as algebra is just a collection of techniques and tips that lots of dedicated algebra puzzlers have developed over many years that can help you solve these algebra puzzles.

Importantly: ***no-one*** can just magically do algebra. Everyone starts off slowly and gets better with practice - just like with any other puzzle. And *everyone* makes mistakes as they go along. If these things describe you, you are ***not*** “*bad at algebra.*” You are learning it.

Variables or Pronumerals

When we learn Algebra, we use letters to represent unknown quantities. We call the letters *pronumerals or variables*. One of the reasons people sometimes find that algebra is hard to get used to is the mixing-up of letters and numbers. But the letters are just an alternative to drawing boxes all the time. The letters represent the unknown quantities. They just indicate a spot that needs to be filled in with a number that makes the equation correct.

For example,

$$36 - \square = 3 \times \square \quad \text{and} \quad 36 - x = 3x \quad \text{and} \quad 36 - a = 3a$$

all describe *equivalent algebraic problems*. We are just using *different symbols to represent the unknown quantities*.

The important point is that;

In any algebraic expression, all instances of a given letter represent the same number.

Expressions and Equations

An ***algebraic expression*** is a phrase that involves variables, numbers and arithmetic operations.

For example,

$$x + 5$$

$$2 \times y - 7 \times x + 3$$

$$a^2 + b^2 - c^2$$

$$x + 2y$$

A ***binomial expression*** is an expression that contains two term such as $x + 5$.

An ***equation*** is a statement of the equality of two mathematical expressions.

For example,

$$x + 5 = 42$$

$$-7x + 3 = x + 2$$

$$a^2 + b^2 - c^2 = 0$$

Operations with Variables

You can do anything with a variable that you can do with a number. After all, a variable is just a place-holder for a number that hasn't been determined yet. So, you can add, subtract, multiply, divide, raise to a power or take a root of a variable in constructing mathematical expressions. Since fractions are just a way of writing division, variables can even appear in fractions.

For the most part we just write the same symbols when we are doing arithmetic with variables as when we are working with numbers. But for multiplication we *almost always* use the convention that when we write two symbols next to each other they are being multiplied together. One reason for this is that the symbols x and X are often used as variables, and they are easy to confuse with $\times$. Also, it's quicker to write expressions without the $\times$. We usually write the numbers first, then the letters in alphabetic order.

Here are some expressions written with the $\times$ symbol and the equivalent expressions written without the $\times$ symbol:

With $\times$	Without $\times$
$3 \times x^2 + 5 \times x \times y$	$3x^2 + 5xy$
$3 \times \left(\frac{x}{5} + 2\right) - 7$	$3\left(\frac{x}{5} + 2\right) - 7$
$\frac{1}{a} + 2 \times \sqrt{x} + 3$	$\frac{1}{a} + 2\sqrt{x} + 3$
$2 \times (x - 2) \times (a \times b - 6)$	$2(x - 2)(ab - 6)$

The following abbreviations are used to write algebraic expressions in their simplest form:

1. Mathematicians ***leave out the multiplication sign*** if a variable is involved.

For example, $x \times y$ is written xy

$3 \times x^2$ is written $3x^2$

$2 \times a \times b$ is written $2ab$

$2a \times 3b \times 4c = 24abc$

However, 2×3 is NOT written 23 as that means something different!

2. Mathematicians ***rarely use the division sign*** $\div$, preferring to use fractions.

For example, $x \div 2$ is written as $\frac{x}{2}$

3. Mathematicians ***usually do not write the number 1 if it is referring to multiplication, the denominator of a quotient or an index.***

For example, $1a = a$, $a^1 = a$, $\frac{a}{1} = a$

Terminology in Algebra

There is some terminology that we need to know. We have names for the various things that can appear in algebraic expressions or equations. It'll seem like a lot of words at first, but you get used to them. We'll discuss them with concrete examples by referring to the following two equations:

$$4y - 7 + 3y \quad \text{and} \quad ax^2 + bx - c = 0,$$

- A ***coefficient*** is a number that multiplies a variable: 4, 3, a and b are coefficients.
- A number on its own is called a ***constant***: 7 and 0 and c are constants.
- A ***term*** is either a single constant or a variable appearing by itself, or product or a fraction involving numbers and variables. The terms in an equation are separated by $+$ and $-$ signs and the $=$ symbol: The terms are $4y$, 7 and $3y$ in the first expression and in the second equation the terms are ax^2 , bx , c and 0.
- ***Like terms*** are terms that have exactly the same variables raised to exactly the same powers. In the first expression, $4y$ and $3y$ are like terms. There are no like terms in the second equation.

Associative and Commutative

To be able to simplify, rearrange, and solve mathematical expressions we use the ***Associative, and Commutative, and Distributive Laws*** because they are fundamental rules that describe consistent patterns in arithmetic and algebra.

Associativity

If we have a string of additions or multiplications, it doesn't matter what order you use to calculate the sums or multiplications. This is called the ***Associative Law***. Examples of the associative law for numbers could be:

$$3 + (4 + 2) = (3 + 4) + 2$$

$$3 \times (4 \times 2) = (3 \times 4) \times 2$$

Since these laws work for all numbers, they also work with pronumerals. In general, the associative law can be written as:

If a , b and c are real numbers, then

$$a + (b + c) = (a + b) + c$$

or

$$a \times (b \times c) = (a \times b) \times c$$

The associative law does not apply to subtraction and division, as seen below.

$$3 - (4 - 2) \neq (3 - 4) - 2$$

$$3 \div (4 \div 2) \neq (3 \div 4) \div 2$$

Commutative

The order of addition or multiplication of numbers does not affect the result. This is called the ***Commutative Law***. An example of the commutative law for numbers could be;

$$3 + 4 = 4 + 3$$

$$3 \times 4 = 4 \times 3$$

Since these laws work for all numbers, they also work with pronumerals. In general, the commutative law can be written as:

If a and b are real numbers, then

$$a + b = b + a$$

or

$$a \times b = b \times a$$

Commutative law does not apply to subtraction and division, as seen below.

$$3 - 4 \neq 4 - 3$$

$$3 \div 4 \neq 4 \div 3$$

Putting commutativity and associativity together, a sum or multiplication of numbers can be calculated in any order.

Key Idea – Standard Sets of Numbers

The set of **Natural numbers**, denoted by $\mathbb{N}$, consists of the **counting numbers**, starting from 1 and is given by:

$$\mathbb{N} = \{1, 2, 3, 4, 5, \dots\}$$

The set of **Integers**, denoted by $\mathbb{Z}$, consist of all **counting numbers, their negatives and 0** and is given by:

$$\mathbb{Z} = \{ \dots, -3, -2, -1, 0, 1, 2, 3, \dots \}$$

Note: *The natural numbers are also called the **positive integers**.*

The set of **Rational numbers** will be denoted by $\mathbb{Q}$. A Rational number is any number which can be expressed as either a **terminating** or **repeating decimal**. Such a decimal can always be expressed as a fraction involving integers.

$$\mathbb{Q} = \left\{ \frac{a}{b} : a, b \in \mathbb{Z} \ b \neq 0 \right\}$$

Examples: $6 = \frac{6}{1},$

$$0.8 = \frac{4}{5},$$

$$0.333... = 0.\dot{3} = \frac{1}{3}$$

$$0.242424... = 0.\dot{2}\dot{4} = \frac{24}{99},$$

$$-2.745 = -\frac{27456}{10000},$$

The **Irrational numbers** are all numbers which are **not rational**. That means that an irrational number cannot be expressed as a quotient of integers.

Examples:

$$\pi = 3.141592653589793238\dots$$

$$e = 2.7182818284590452353\dots$$

$$\sqrt{2} = 1.414213562373095\dots$$

***Note:** The dots indicate that the representation is incomplete, because the exact value cannot be written.*

***Note:** Since we cannot easily work with the non-terminating or non-repeating decimal, irrational numbers are normally approximated by terminating decimals. For example, e can be approximated correct to 4 decimal places by the rational number 2.7183.*

The set of **Real Numbers** is denoted by $\mathbb{R}$ and includes the natural numbers, the integers, the rational numbers and the irrational numbers.

Key Idea - Substitution

While we often use letters to represent numbers, sometimes we are given the value of each variable in an algebraic expression. We can then use substitution to evaluate the expression.

Substitution is the process of **replacing variables by assigned numbers in an algebraic expression**. This allows us to calculate solutions to multiple known values using the same rule, as demonstrated below, which is a powerful application of Algebra.

Example:

If $x = -3$ evaluate $3x^2 - 2x + 7$.

$$\begin{aligned} 3x^2 - 2x + 7 &= 3 \times (-3)^2 - 2 \times (-3) + 7 \\ &= 3 \times 9 - 2 \times (-3) + 7 \\ &= 27 - 2 \times (-3) + 7 \\ &= 27 + 6 + 7 \\ &= 40 \end{aligned}$$

Note: Remember to follow the correct order of operations when evaluating expressions

Example:

The rule for converting a temperature expressed in Celsius scale (C) to one expressed in Fahrenheit scale (F) is: $F = \frac{9C}{5} + 32$. Convert 50°C , -20°C and 0°C into Fahrenheit.

$$F = \frac{9 \times 50}{5} + 32 = 122^{\circ}\text{F}$$

$$F = \frac{9 \times (-20)}{5} + 32 = -4^{\circ}\text{F}$$

$$F = \frac{9 \times 0}{5} + 32 = 32^{\circ}\text{F}$$

Examples: If $m = 6$, $n = -3$, find the value of $\sqrt{5m - 2n}$

$$\begin{aligned}\sqrt{5m - 2n} &= \sqrt{5 \times 6 - 2 \times (-3)} \\ &= \sqrt{36} \\ &= 6\end{aligned}$$

Examples:

Substitute $T = 2.5$, $g = 9.8$ into $L = g\left(\frac{T}{2\pi}\right)^2$ to find the length L of a pendulum.

$$\begin{aligned} L &= g\left(\frac{T}{2\pi}\right)^2 \\ &= 9.8 \times \left(\frac{2.5}{2\pi}\right)^2 \\ &= 1.55 \text{ (2 decimal places)} \end{aligned}$$

Examples:

What is the mistake in the working below, when $x = -2$ is substituted into $x^2 + x - 1$?

$$\begin{aligned} -2^2 + -2 - 1 &= -4 - 2 - 1 \\ &= -7 \end{aligned}$$

The mistake is with the **missing brackets when substituting a negative number**

It should be written as,

$$\begin{aligned} (-2)^2 + -2 - 1 &= (-2)^2 - 2 - 1 \\ &= 4 - 2 - 1 \\ &= 1 \end{aligned}$$

Key Idea - Simplification

In *algebra*, *like terms* are terms that have *exactly the same variable or combination of variables*, each raised to the *same power*. This means the variable part of the terms must be identical — the coefficient (numbers in front) of the variables can be different.

Example,

- $3x$ and $7x$ are like terms because both have the variable x
- $2a^2b$ and $-5.32 a^2b$ are like terms because both contain a^2b
- 44 and -9 are also like terms because they are both constants (no variables).

However, terms such as $2x^2$ and $2x$ are **not** like terms because the powers of x are different. Similarly, $3x$ and $3y$ are not like terms because they have different variables.

Discussion Question:

Which of the terms below are “like terms”?

$$2y$$

$$-14y$$

$$-3abc^2$$

$$\frac{1}{3}y$$

$$0.5a^2b$$

$$abc^2$$

$$\frac{x}{2}$$

$$5.2x$$

$$kx$$

$$\frac{-2}{3}abc^2$$

$$a^2b$$

Addition and Subtraction of Algebraic Expression

Like terms can be added or subtracted in order to simplify expressions.

Examples:

Simplify the following if possible:

$$\begin{aligned}5x + 3y - z - 2x + 4y + 2z &= 5x - 2x + 3y + 4y - z + 2z \\ &= 3x + 7y + z\end{aligned}$$

$$\begin{aligned}6ab + a^2 - 3b^2 + 3ab - 0.5a^2 &= 6ab + 3ab + a^2 - 0.5a^2 - 3b^2 \\ &= 9ab + 0.5a^2 - 3b^2\end{aligned}$$

$$\begin{aligned}(-x^2y + 3x^2) - (xy^2 - x^2) &= -x^2y + (3x^2 + x^2) - xy^2 \\ &= -x^2y + 4x^2 - xy^2\end{aligned}$$

$$3x^3 - 2x + 3xy - 6 \quad \text{This expression cannot be simplified as there are no like terms!!}$$

Note:

The minus sign belongs to the term immediately after it.

Multiplication of Algebraic Expressions

When **multiplying** two or more algebraic expressions containing numbers and variables, **multiply the numbers together and write the variables without any multiplication signs.**

Repeated multiplication of variables is written as indices, such as $a \times a \times a = a^3$. We will look at indices in more detail later in this subject.

Examples:

Simplify the following:

$$2a \times -3a = -6a^2$$

$$\begin{aligned} 9xy \times 5yz &= 9 \times 5 \times xy yz \\ &= 45xy^2z \end{aligned}$$

$$\begin{aligned} 3xy^2 \times -2x^3y &= 3 \times -2 \times x \times x^3 \times y^2 \times y \\ &= -6x^4y^3 \end{aligned}$$

$$-2a^2 \times 3b \times -2ab = 12a^3b^2$$

Division of Algebraic Expression

When **simplifying fractions**, only **factors that appear in every single term in the fraction can be simplified** (cancelled out).

Examples:

Simplify the following:

$$\frac{42ab}{6ac} = \frac{7b}{c}$$

$$\begin{aligned}\frac{4m^2 \times 5np \times 3p}{2m \times 10p^2 \times 6} &= \frac{60m^2np^2}{120mp^2} \\ &= \frac{mn}{2}\end{aligned}$$

$$\begin{aligned}\frac{9xy \times (-4x)}{-2x \times 3y} &= \frac{3 \times 3 \times x \times y \times (-4 \times x)}{-2 \times x \times 3 \times y} \\ &= \frac{3 \times (-4 \times x)}{-2} \\ &= -6x\end{aligned}$$

Exercise: Simplify the following:.

$$(a) \quad \underline{3ab} - \underline{a^2} + \underline{3b^2} - \underline{a} - \underline{3a^2} + \underline{2ab} = 3ab + 2ab - a^2 - 3a^2 + 3b^2 - a \\ = 5ab - 4a^2 + 3b^2 - a$$

$$(b) \quad -4a \times 3a^2b \times b = -12 \times a \times a^2 \times b \times b \\ = -12a^3b^2$$

$$(c) \quad (-xy) \times 2xz \times xz = -2 \times xy \times xz \times xz \\ = -2x^3yz^2$$

$$(d) \quad \frac{6y^2 \times 7y}{xy^2 \times 7xy \times 3y} = \frac{42y^3}{21x^2y^4} = \frac{42 \times \cancel{y} \times \cancel{y} \times \cancel{y}}{21 \times \cancel{x} \times \cancel{x} \times \cancel{y} \times \cancel{y} \times \cancel{y} \times \cancel{y}} \\ = \frac{2}{x^2y}$$

Key Idea - Expanding using the Distributive Law

Expansion means **multiplying out the** product of two or more algebraic expressions, each of which contains one or more terms. Expansion is also known as “**removing brackets**”.

Discussion Question:

How do we calculate 5×18 mentally?

$$5 \times (10 + 8) = 50 + 40 \\ = 90$$

Can you write a general form for this expansion?

$$a(b + c)$$

Distributive Law involves multiplying the term outside the brackets by each term inside. In the example below note that, when the brackets are removed, both terms in the brackets are multiplied by a .

$$a(b \pm c) = a \times b \pm a \times c$$

or

$$a(b \pm c) = ab \pm ac$$

Example,

Expand and simplify:

$$\begin{aligned} 2(x - 5) + 3x &= 2 \times x - 2 \times 5 + 3x \\ &= 2x - 10 + 3x \\ &= 2x + 3x - 10 \\ &= 5x - 10 \end{aligned}$$

$$\begin{aligned} 5ab + b(6a - 7) &= 5ab + b \times 6a - b \times 7 \\ &= 5ab + 6ab - 7b \\ &= 11ab - 7b. \end{aligned}$$

Note: A negative sign in front of a set of brackets represents the entire expression in brackets being multiplied by (-1) .

$$\begin{aligned} -(b + c) &= (-1)(b + c) \\ &= (-b) + (-c) \\ &= -b - c \end{aligned}$$

For example,

$$\begin{aligned} -(x + y^2 - z) &= (-1)(x + y^2 - z) \\ &= (-x) + (-y^2) - (-z) \\ &= -x - y^2 + z \end{aligned}$$

Exercise: Expand and simplify each of the following expressions:

(a) $3(x + 4) - 6x$  $= 3x + 12 - 6x$
 $= -3x + 12$

(b) $\frac{1}{x}(x^2)(x+1) - x(x-1)$ 
 $\frac{1}{x}(x^2) = \frac{1}{x} \times \frac{x^2}{1}$
 $= \frac{x^2}{x}$
 $= x$
 $= x(x+1) - x(x-1)$
 $= x^2 + x - x^2 + x$
 $= 2x$

(c) $4x + y + 2z - 3(x + y^2 - z)$ 
 $= \underline{4x} + y + \underline{2z} - \underline{3x} - 3y^2 + \underline{\underline{3z}}$
 $= x + 5z + y - 3y^2$